

Location Services Feature on Mobile Devices

Overview

When OpenPath RPI is opened on a mobile device, the web app should have the ability to use location services to mark the user's realtime location. This feature will be used to open the map at the user's current location, and move the map dynamically with the user during navigation mode. ***This feature should be implemented once navigation is functional, and a responsive, mobile version of the website exists and is accessible.***

Mobile Website vs Standalone App Approach

Mobile Website (Opened in Phone Browser)	Standalone App (iOS, Android)
• Recommended • Location permission must be provided by user each time the OpenPath website is opened on a phone • Location can still be tracked, and used to pan and zoom the map dynamically • Much easier implementation: same backend, frontend only has minor scaling and element changes to match screen size	• Requires development of an entire mobile client (React Native) • Location Services can be tracked in real time, even in background • Custom push notifications for navigation can be created (ex. Google Maps, DoorDash style notifications) • Not feasible within F26

Specifications

Within the backend, there must be a check for the user's device type. When a mobile device user is detected, load the mobile version of the website for the frontend. It will be the same website, same data, but elements are scaled appropriately for a phone or tablet screen. Then, the system must ask the user for location services.

- If the user declines, the navigation paths will still show, and the map will still work, however, the map will not be able to dynamically follow the user's location.
- If the user accepts, the map will pan and zoom to their current location, display an arrow at their location. When a navigation mode is started, the map will pan, zoom, and rotate in real time following the user's location as they walk along the path. (Similar to Google Maps on a phone, or a mini map system in a video game).